

Experiment No. 23

Title

Prolog Program to Implement Family Tree

Aim

To write a Prolog program to represent a **family tree** and find family relationships.

Objective

The objective of this experiment is to understand how Prolog facts and rules can represent family relationships and answer queries.

Algorithm

1. Start.
2. Define parent facts.
3. Define rules for family relationships.
4. Execute a query.
5. Display the relationship.
6. Stop.

Flowchart

Start

|

v

Define Family Facts

|

v

Define Rules

|

v

Execute Query

|

v

Find Relationship

|

v

Display Result

|

v

Stop

Prolog Program

```
parent(john, mary).
```

```
parent(john, david).
```

```
parent(mary, susan).
```

```
parent(mary, robert).
```

```
father(X,Y) :-
```

```
    parent(X,Y).
```

```
grandparent(X,Z) :-
```

```
    parent(X,Y),
```

```
    parent(Y,Z).
```

Queries

```
?- father(john,mary).
```

Output:

```
true.
```

```
?- grandparent(john,susan).
```

Output:

```
true.
```

Result

The Prolog program successfully represented the family tree and retrieved family relationships using queries.

Conclusion

Thus, the family relationships were successfully implemented and queried using Prolog.